Beyond

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Beyond, a tool that helps put inclusive and accessible design into action

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TABLE OF CONTENTS

ACKNOWLEDGMENTS 6

ABSTRACT 7

1. INTRODUCTION 8
- 1.1 Motivation 9
- 1.2 Goals 10
  - 1.2.1 Problem Areas to Solve 10
  - 1.2.2 Desired Outcomes 11
- 1.3 Target Demographic 12
- 1.4 Problem Statement 12

2. RESEARCH 15
- 2.1 Secondary Research 16
  - 2.1.1 Statics and Numbers 16
  - 2.1.2 Existing Results and Study 20
- 2.2 Primary Research 22
  - 2.2.1 User Interview 22
- 2.3 Key Findings 24
- 2.4 Assumption Mapping & Hyphothesis 26
- 2.5 Competitor Analytics 28
- 2.6 Value Proposition 30

3. DESIGN 31
- 3.1 Ideation 32
  - 3.1.1 Ideas + Criteria metrics 32
  - 3.1.2 Features and Content for MVP 34
- 3.2 User Journey 36
  - 3.2.1 5E's Experience Mapping 36
- 3.3 Information Architecture 39
- 3.4 User Testing with Initial Prototypes 40
- 3.5 Key Concepts 42
- 3.6 Key Features and Content 44
  - 3.6.1 Barrier Awareness 44
  - 3.6.2 Interface Design Assistant 46
  - 3.6.3 Annotation Assistant 52

4. CONCLUSION 62
- 4.1 Final Thoughts 63
- 4.2 Future Work 64

REFERENCE 66

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Lastly, I would like to express my appreciation to Eric Forman who has guided and encouraged me to pursue my academic goals.

ABSTRACT

This thesis project presents a comprehensive solution to enhance inclusive and accessible design, Beyond. Beyond is a tool that leverages automated audits and seamless education to help designers implement accessibility measures before development in their work with less effort.

Beyond also provides an accessibility mode that verifies the quality of accessibility annotations such as focus order, landmarks, and alt text by simulating them in an assistive technology environment. This mode allows designers to test the accessibility of their work using a keyboard and screen reader, ensuring that it is truly inclusive and accessible for all users.

What sets Beyond apart is its support for accessibility testing with people who have disabilities. By involving these users in the testing process, Beyond ensures that design and development efforts are truly inclusive and provide a seamless experience for everyone.

To validate the efficacy of Beyond, a prototype was tested with a group of designers. The results showed that Beyond is an effective solution for enhancing inclusive and accessible design, and that involving users with disabilities in the testing process is critical for ensuring the success of accessibility efforts.

Overall, Beyond is a powerful tool that helps put inclusive and accessible design into action, with less effort, and in a way that truly benefits all users, including those with disabilities.

INTRODUCTION

1.1 Motivation
1.2 Goals
1.2.1 Problem Areas to Solve
1.2.2 Desired Outcomes
1.3 Target Demographic
1.4 Problem Statement

1 INTRODUCTION

1.1 Motivation

The motivation behind this thesis stems from my personal journey in discovering the importance of Inclusive Design. It began in 2018 when I created a poster for the community in Chicago and realized the need to involve diverse groups of people in human illustrations. This experience sparked my interest in inclusive design for digital products, but I soon realized that it was difficult to practice in a homogeneous society like Korea.

As a result, I came to New York, a city known for its diversity, to study inclusive design for diverse people.

It wasn't until I interned at Google and worked on redesigning the 'Recently viewed' section that I truly understood the importance of inclusive design. The UX writer's feedback on how not everyone can "view" the page but instead may "hear" it through a screen reader was a wake-up call for me. It made me realize that good intentions alone were not enough; I needed to actively practice inclusive design.

Since then, I learned accessible and inclusive design. However, even with this knowledge, I found that creating truly inclusive designs was not an easy task.

Through my experiences, I discovered that many product designers struggle with inclusive and accessible design despite having good intentions. This motivated me to delve deeper into the subject and explore how designers can effectively implement inclusive design practices in their work.

Therefore, the purpose of this thesis project is to provide actionable and motivating solution for product designers to create more inclusive designs. By sharing my personal journey and experiences, I hope to inspire others to embrace inclusive design and make a positive impact on the lives of diverse users.

1.2 Goals

1.2.1 Problem Area to Solve

Inaccessibility in digital design: Many digital designs and applications lack accessibility features, which can make it difficult or impossible for people with disabilities to use them. According to the WebAIM Million report, only 2.2% of the sampled web pages met the most basic accessibility guidelines.

Limited support for accessibility testing with people with disabilities: In many cases, accessibility testing is done solely by designers and developers, without input from people with disabilities. This can lead to an incomplete understanding of accessibility issues and can result in designs that are not truly inclusive. According to the World Health Organization, around 15% of the world's population, or about 1 billion people, have a disability.

Lack of automated accessibility tools: While there are some automated accessibility tools available, they often require significant manual effort and expertise to use effectively. This can be a barrier for designers and developers who may not have the time or resources to manually check for accessibility issues. UserZoom's study shows that only 29% of accessibility issues are caught by automated accessibility testing, while the remaining 71% require manual testing by experts.

Quality of accessibility annotations: The quality of accessibility annotations, such as alt text for images, can vary widely and can significantly impact the accessibility of a design. Without a way to verify the quality of these annotations, it can be difficult to ensure that a design is truly accessible. In a survey by UserZoom, 71% of respondents with disabilities reported encountering barriers when using websites or apps.

Lack of inclusive content: Exclusive content, such as images and UX copy, can exclude marginalized people in digital world. For example, stock images that only feature people with a certain skin color or body type can perpetuate harmful stereotypes and exclude people from diverse backgrounds. Similarly, UX copy that assumes a certain gender or cultural background can make users from diverse backgrounds feel excluded or marginalized.

1.2.2 Desired Outcomes

Beyond aims to make it easier for designers to create inclusive and accessible designs, while also ensuring that people with disabilities are involved in the testing process and that the quality of accessibility annotations is verified.

1.3 Target Demographic

The primary audience for Beyond is designers who are responsible for creating digital design. According to a survey by Springboard, there are over 1 million digital product or UX/UI designers worldwide.

Beyond aims to make it easier for designers to create inclusive and accessible designs that meet WCAG 2.1 standards at a minimum. By providing an automated auditing tool and education resources, designers can easily integrate accessibility into their design process, without requiring extensive knowledge of accessibility guidelines. Moreover, Beyond's accessibility mode enables designers to test and check the quality of their accessibility annotations. Conducting accessibility testing with people with disabilities ensures that designs are truly accessible to all users.

1.4 Problem Statement

In today's digital landscape, it is critical to create inclusive and accessible digital experiences. However, designing for accessibility and inclusivity can be complex and time-consuming. It often requires extensive knowledge of accessibility guidelines and manual auditing processes. As a result, many designers may overlook accessibility and inclusivity considerations, leading to inaccessible designs that exclude marginalized individuals.

Problem Statement

How might we make inclusive and accessible design in the interface design process more actionable and motivating?

Photo by iStock

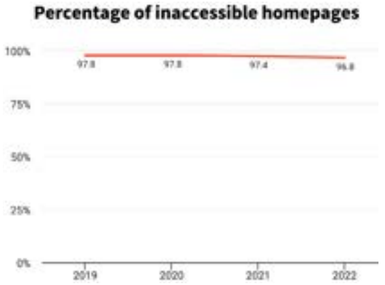
RESEARCH

2.1 Secondary Research
2.1.1 Statics and Numbers
2.1.2 Existing Results and Study
2.2 Primary Research
2.2.1 User Interview
2.3 Key Findings
2.4 Assumption Mapping & Hyphothesis
2.5 Competitor Analytics
2.6 Value Proposition

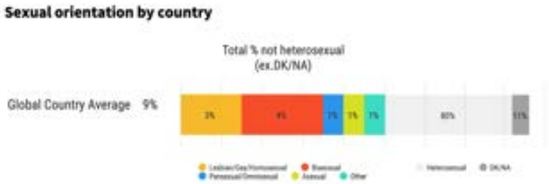
2 RESEARCH

2.1 Secondary Research

2.1.1 Statics and Numbers



Source: The WebAIM Million, 2022



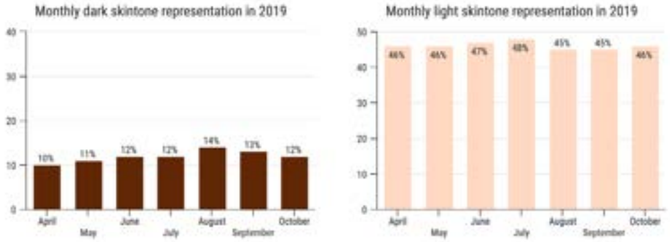
Source: LGBT+ Price 2021 Global Survey, Ipsos

Disability-inclusion

WHO reports 15% of people (1.2 billion) live with disabilities, but web accessibility remains low. Evaluations of top 1million websites show very little improvement in the past 4 years: 97.8% had issues in 2019/2020, 97.4% in 2021, and 96.8% in 2022.

LGBTQIA-inclusion

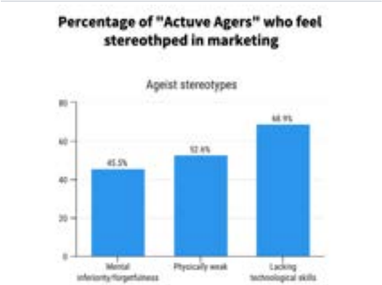
A survey of people in 27 countries found that 9% of people worldwide (720 million) identify on the LGBTQIA spectrum, and perhaps some of the 11% who responded, “don’t know’ or “won’t say” are in the spectrum as well. 71% of LGBTQIA consumers are more likely to engage with an ad that represents their sexual orientation.



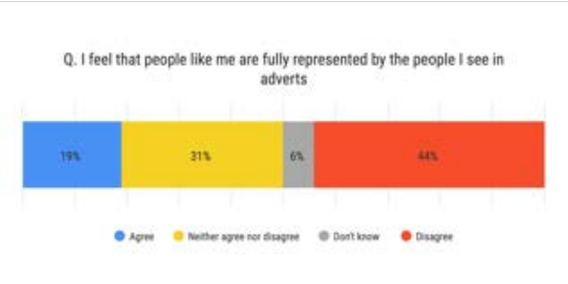
Source: Beauty brands were quick to tout diversity during summer protests, but the commitment was short-lived, Fast Company, 2020

Racial/Ethnic-inclusion

The global prioritization of light skin tones is a significant issue that disproportionately affects Black, African, Asian Indigenous, and Latino communities, who make up the majority of the world’s population at 80% (6.4 billion). This discrimination is evident in the lack of diversity in ad campaigns, as seen in a study by Eyecue Insights of 70 beauty brands. The study found that only 10% of beauty images featured dark skin tones, while about 50% featured light skin tones.



Source: The WebAIM Million, 2022



Source: Diversity in ads: Are brands doing enough, YouGov, 2021

Age-inclusion

In 2022, people aged 65 and older make up 10% (800 million) of the world population, according to the United Nations. In the United States, 75 million people aged 55 and over are considered "Active Agers," who are mentally, socially, and digitally active. Of these Active Agers, 66% agree they are stereotyped in marketing.

Across 17 markets, more than two-fifths of global consumers (44%) feel that they are not fully represented by the people they see in adverts.

Web accessibility remains a significant issue as 97% of homepages are still inaccessible. Additionally, 44% of global consumers feel that they are not fully represented in advertisements.

2.1.2 Existing Results of Study

1. The involvement of people with disabilities in the design process can significantly improve the inclusivity and accessibility of digital products.

A study conducted by the University of Michigan found that by involving people with disabilities in the design process, designers and developers gain valuable insights into the needs and challenges faced by users with disabilities. As a result, the study concluded that incorporating the feedback and suggestions of people with disabilities leads to the creation of websites that are more usable and accessible to all users, regardless of their abilities.

2. Designing for accessibility from the beginning of a project can save time and resources in the long run.

The International Association of Accessibility Professionals (IAAP) conducted a study that found that by incorporating accessibility from the outset of a project, designers and developers can avoid the need for costly retrofits and redesigns later on. By prioritizing accessibility from the beginning, designers can ensure that the website or digital product is built with inclusivity and accessibility in mind, leading to a better user experience for all.

3. Providing accessibility training for designers and developers can improve the implementation of accessible design.

A study conducted by the National Center on Disability and Access to Education found that providing accessibility training to designers and developers can lead to a significant improvement in their understanding and implementation of accessibility guidelines. The study concluded that by equipping designers and developers with the knowledge and tools to create accessible digital products, they are better able to create inclusive designs that meet the needs of a diverse range of users.

4. Regular accessibility testing and evaluation is key to identifying and addressing accessibility issues in digital products.

A study conducted by the University of Wisconsin-Madison found that regular accessibility testing and evaluation led to a significant improvement in the accessibility of websites over time. By conducting regular accessibility audits, designers and developers can identify areas where improvements are needed and make the necessary changes to ensure that the digital product is accessible to all users.

5. Including diverse images and avoiding stereotypes are important steps towards creating a more inclusive and welcoming environment.

A study conducted by Creative Review found that by including diverse images in advertising, designers and marketers can help to create a more inclusive environment that reflects the diversity of their target audience. The study also found that avoiding harmful stereotypes and biases can help to promote inclusivity and prevent harm to marginalized groups. By showcasing a range of diverse images and avoiding stereotypes, designers and marketers can help to create a more inclusive and welcoming environment for all.

6. Inclusive language and avoiding microaggressions are crucial components of creating an inclusive and respectful environment.

A study conducted by the University of California, San Francisco found that using inclusive language and avoiding microaggressions in communication can help to create a more inclusive and respectful environment. Inclusive language refers to using language that is gender-neutral, non-offensive, and avoids assumptions about a person's gender or identity. Microaggressions, on the other hand, refer to subtle but harmful comments or actions that can perpetuate stereotypes and biases. By using inclusive language and avoiding microaggressions, individuals can help to create a more inclusive and respectful environment that values diversity and promotes inclusivity.

2.2 Primary Research

2.2.1 User Interview

The purpose of conducting user interviews for my thesis project is to gain a deeper understanding of designers' work processes and their current practices for inclusive and accessible design. By engaging with designers directly, I aim to learn about their personal experiences and perspectives on inclusive design, as well as the most challenging aspects of incorporating these principles into their work.

I conducted user interviews with a total of eight UX designers, consisting of four senior-level and four junior-level designers. These individuals were selected based on their experience and expertise in the field of UX design. Through these interviews, I was able to gather a range of perspectives on the challenges and opportunities associated with inclusive design.

[Screening]

- For how long have you been working as a product/UX/UI designer?
- What is the size of your company?
- How many designers are on your design team?

[Questions]

- On a scale of 0-10, how would you rate yourself as a designer when it comes to incorporating inclusive design in your work?
- At what stage of the design process do you consider accessibility?
- What does inclusive design mean to you personally?
- Have you taught yourself about inclusive design?
- Have you learned about inclusive design in an academic setting?
- Do you think your work in inclusive design is adequate? What areas do you think need improvement?

Quotes from Interviews

"I think designers should try to use the screen reader. "

"It was a disappointment that I couldn't conduct usability tests with individuals with disabilities."

"I hope there is something that I can check accessibility or inclusion quickly."

"I think time is the biggest problem. I want to put effort into this, but I feel like I don't have enough time to do it."

"I've looked for a multicultural design. I've only recently become aware of this topic and realized that it's something I need to consider."

"I have read articles about accessibility and inclusive design on Nelson and Medium. I wanted to verify if my understanding is correct."

"When creating a text input form, I referred to designs from other major companies like IBM and Microsoft. I considered them as benchmarks for inclusive design and made sure that they were accessibility compliant."

"Despite running out of time, I'm striving to make my design as accessible as possible. Failing to do so would disappoint me, as I would feel like I hadn't fulfilled my responsibility. At the very least, I am checking the color contrast."

"As my company has an accessibility team, it seems to be implementing it well functionally. However, inclusion efforts are not performing as well, and it is difficult to evaluate their success."

"I'm interested in inclusive and accessible design, but it's not easy to apply when I work. I don't know where to start studying. I hope there is something that I can check accessibility or inclusion quickly."

"I totally think it's super important for every designer to dedicate time to inclusive and accessible design. Recognition is the first step. In school, we're taught about it, but after graduation, no one emphasizes its importance. Designers have to actively seek out opportunities to learn and work in such an environment."

"I want to learn how screen readers work and review a checklist that ensures accessibility."

2.3 Key Findings

1

Involving people with disabilities in the design process ensures genuine accessibility

2

Exclusion can be difficult to detect, even with good intentions

3

Designers are overwhelmed by the extensive knowledge required for inclusive and accessible design

4

Designers need motivation to take action

2.4 Assumption Mapping & Hyphothesis

To properly test all of the assumptions that may arise in a given situation, it can be overwhelming to try to tackle them all at once. A great way to approach this is by utilizing a Prioritization Matrix. This tool can help identify which assumptions should be tested first, based on the level of risk associated with each and the number of unknown variables. In this way, one can focus their efforts on the most critical and impactful assumptions, increasing the likelihood of success overall.

In Figure 1.1, I have outlined my assumptions regarding inclusive and accessible design practices. These assumptions are based on primary research and experience. Therefore, my hypothesis for this project is that digital product designers will engage in inclusive and accessible design when provided with an actionable and motivating tool.

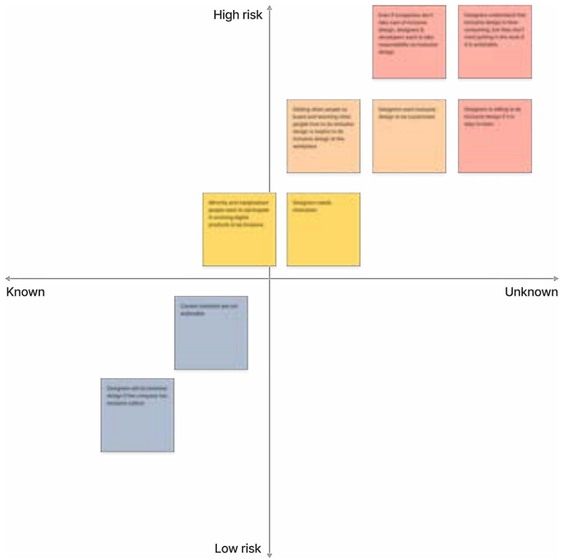


Figure 1.1 Assumption apping for Beyond

Hypothesis

I believe that if implementing inclusive and accessible design is actionable and motivating, digital product designers will practice it.

2.4 Competitor Analytics

Competitor analytics is a crucial process for evaluating the performance and strategies of competitors in a particular market or industry. One effective tool for conducting a competitor analysis is the Competitor Analysis Matrix. This matrix consists of a grid that allows for a comparison of the strengths and weaknesses of a company's products or services with those of its competitors.

Figure 1.2 depicts the competitors and their offerings in each feature of Beyond. It is important to note that while competitors may serve a part of the accessible and inclusive design process, Beyond covers the entire process, as illustrated in Figure 1.3. This analysis ensures Beyond stands out in the market and provides unique value to users.

Category	Competitors	Introduction
Accessibility	Stark	Stark is a suite of integrated accessibility tools. It consists of a plugin for design tools like Figma and Sketch. In addition to color contrast checking, Stark also offers tools for checking font sizes and line spacing, as well as for generating color palettes that meet accessibility guidelines.
	Accessibe	AccessiBe is a cloud-based platform that uses AI and machine learning to automatically improve website accessibility. It provides a wide range of features, including an accessibility interface, AI-powered accessibility enhancements, continuous monitoring and updates, compliance audit, keyboard navigation, and ARIA compatibility. It is focused on helping website owners make their existing websites more accessible.
Inclusive content	Textio	Textio is an augmented writing platform that can give your team a score on the content they write. Plus, it provides suggestions on how to improve. This includes bias interruption, expanded language insights, and team analytics. The whole point is to help you write more insightful, inclusive content.
	Witty	Witty is a digital writing assistant that supports inclusive language in all company communications. Integrated in your workflow, it operationalizes inclusion.
Accessibility Mode	Figma Accessible prototype	Accessible prototypes in Figma allow you to navigate through content using a screen reader. To use accessibility mode in a prototype, you'll first need to install a supported screen reader. Accessible prototypes are only available in the desktop browser and on the Figma desktop app. Accessibility mode does not work for mobile or mobile web prototypes.
Accessibility testing	AccessibilityOz	AccessibilityOz is a digital accessibility consulting firm that offers a range of services, including user testing with people with disabilities. They specialize in testing with people who use assistive technologies, such as screen readers, voice recognition software, and keyboard navigation.
	Userlytics	Userlytics is a user testing platform that offers testing with people with disabilities. They have a panel of testers with various disabilities, including vision, hearing, motor, and cognitive impairments.

Similar Areas/Competitors	Integrability with design tool	Educational	Cultivating empathy	Functionality with prototype	Assessment and suggestions	Involving people with disabilities	Easy to use	Accessibility standards checking	Accessible components library	Inclusive content checking	Inclusive content library
Beyond (My solution)	✅	✅	✅	✅	✅	✅	✅	✅	✅	✅	✅
Accessibility											
Stark	✅	✅	✅		✅		✅	✅			
AccessiBe		✅	✅		✅		✅	✅			
Inclusive content											
Witty		✅			✅		✅			✅	
Textio		✅			✅		✅			✅	
Accessibility Mode											
Figma-present	✅	✅	✅	✅							
Accessibility testing											
AccessibilityOz						✅	✅				
Userlytics	✅				✅	✅	✅				

Figure 1.2 (Left) List of competitors of Beyond

Figure 1.3 (Top) Competitor analytics matrix

Value Proposition

Beyond is a design tool that helps designers achieve inclusivity and accessibility in digital services through automatic assessments and suggested corrections.
It is an empathy-driven solution that provides a unique experience of involving people with disabilities to go beyond accessibility standards.

DESIGN

3.1 Ideation
3.1.1 Ideas + Criteria metrics
3.1.2 Features and Content for MVP
3.2 User Journey
3.2.1 5E's Experience Mapping
3.3 Information Architecture
3.4 User Testing with Initial Prototypes
3.5 Key Concepts
3.6 Key Features and Content
3.6.1 Barrier Awareness
3.6.2 Interface Design Assistant
3.6.3 Annotation Assistant

3 DESIGN

3.1 Ideation

3.1.1 Ideas + Criteria metrics

Figure 2.1 illustrates the decision-making process to help objectively evaluate different options based on a set of predetermined criteria or metrics. The matrix helps identify which ideas meet the most criteria, but personal interests should also be considered. The criteria metrics serve as a helpful guide throughout the thesis project.

Throughout the framework, accessibility features and education, inclusivity in content features and education, assistive technology simulator, and user testing with people who have disabilities are given the highest priority.

			♥	♥	♥		♥		
	Roadmap/ Progress bar	Narratives of challenges	A11y feature and education	Inclusivity in content feature and education	Assistive Technology Simulator	Learn WCAG with visuals	User testing with real people	My activities/ peers' activities	Annual report
User's needs									
Save time			v	v		v	v		
Less effort to learn			v	v	v	v			
Less effort to design inclusively			v	v	v		v		
Find problems regarding A11y and inclusivity			v	v	v		v		
Engaged	v	v			v		v	v	v
Verify if design is accessible indeed					v		v		
Understand A11y			v		v	v	v		
Learn inclusivity deeper			v	v			v		v
Personal criteria									
Content-focused		v	v	v		v			v
Love-able	v	v	v	v	v	v	v	v	v
Humane: socially-responsible		v	v	v	v	v	v	v	v
Ethical: alongside my morals and beliefs		v	v	v	v	v	v	v	v
Knowledge of font-end development			v	v	v	v			
Useful	v		v	v	v	v	v	v	v
Viable in the future	v	v	v	v	v	v		v	v
TOTALS	4	6	13	12	11	10	11	6	8

Figure 2.1 Ideas + Criteria metrics for Beyond

3.1.2 Features and Content for MVP

The minimum viable product (MVP) is the simplest version of a project that can still provide value to users. The MVP is developed early on in the project and serves as a baseline for testing and further development. Typically, the MVP includes only the core features and functionality necessary to demonstrate the project's value proposition. It may lack some of the advanced features and polish of the final product, but it should still provide a meaningful and useful experience for users.

Figure 2.2 illustrates the core content to explore for the MVP of Beyond, while Figure 2.3 shows the core features to develop for the MVP. Prioritized features are evaluated based on their importance, ease of implementation, and urgency.

Category	Content	Touch point	Relevant feature	MVP?
Accessibility	**Color contrast**	Figma Plugin	Accessibility audit	v
	Typography	Figma Plugin	Accessibility audit	
	Touch target	Figma Plugin	Accessibility audit	
	Alt-text	Desktop app	Annotation assistant	v
	Focus order	Desktop app	Annotation assistant	
	Landmark	Desktop app	Annotation assistant	v
	Multimedia	Desktop app	Annotation assistant	
Content Inclusivity	**UX Copy**	Figma Plugin	Content inclusivity audit	v
	Images of People	Figma Plugin	Content inclusivity audit	
	Gender identity form	Figma Plugin	Content inclusivity library	v
	Assigned Sex at Birth	Figma Plugin	Content inclusivity library	
	Race form	Figma Plugin	Content inclusivity library	
	Name form	Figma Plugin	Content inclusivity library	
	Marital Status	Figma Plugin	Content inclusivity library	
	Input form	Figma Plugin	Accessibility library	
	Date picker	Figma Plugin	Accessibility library	
	Button	Figma Plugin	Accessibility library	

Figure 2.2 Content for MVP

Category	All the features I want to make	Value to provide / the problems it will solve	Important (1-3)	Ease (1-3)	Urgency (1-3)	Status	MVP?	Touchpoint
Login	SSO with Figma account	Easy to access and connect to design files	3	1	1	Need exploration	v	Desktop app
Design	Content Inclusivity audit & library	Save time and effort for inclusive design	3	3	3	Need exploration	v	Figma Plugin
Design	Accessibility audit & library	Save time and effort for inclusive design	3	3	2	Need refinement	v	Figma Plugin
Design	Integrating features and education - Accessibility	Save time and effort for inclusive design	3	3	2	Need refinement	v	Desktop app
Learn	Learning deeper - visual version of WCAG	Reduce cognitive effort to understand a11y	1	2	1	Need exploration		Desktop app
Onboarding	Empty status	Help designers to adjust the system and start working	1	2	1	Need exploration	v	Desktop app
Motivation	Voice of marginailzed groups of people (Videos and audio)	Motivate designers to design inclusively and empathize difficulty of marginalized people	3	2	2	Need exploration	v	Desktop app
Motivation	Progress bar	Encourage designers to practice inclusive design.	1.5	1	2	Need refinement		Desktop app
Motivation	Peers' recent activities (+my recent activities)	Give designers a light but great nudge to keep doing inclusive design (positive peer pressure)	2.5	1	1	Need refinement		Desktop app
Motivation	Roadmap (Generating to-do list with expected time)	Give a nudge to designers to start and feel less-overwhelmed	1	1	1	Need exploration		Desktop app
~~Motivation~~	~~Badges~~	~~Engage designers~~	~~scope out~~	~~1~~		~~Need exploration~~		~~Desktop app~~
Testing	Assistive Technology Simulator	- Help designers empathize more with the difficulties of people using AT - Prevent waste of development resources	2	2	3	Need exploration	v	Desktop app
Testing	User testing with real people (people with disabilities)	- Streamline the complicated process of testing their prototypes with people with disabilities	3	2	3	Need exploration	v	Desktop app
~~Reflection~~	~~Annual report~~	~~Help designers reflect on their strengths and weakness in inclusive design~~	~~scope out~~	3		~~Need exploration~~		~~Email/notification~~
Handoff	Export/Sync guidelines	Reduce the complexity of various channels to collaborate with developers	2	2	2	Need exploration	v	Desktop app/Figma
~~Handoff~~	~~Code for developers~~	~~Expedite collaboration between designers and developers~~	~~scope out~~	~~2~~		~~Need refinement~~		~~Desktop app~~

Figure 2.3 Features for MVP

3.2 User Journey

3.2.1 5E's Experience Mapping

5E mapping is a framework for designing and organizing learning experiences. The 5E model consists of five phases: Engage, Explore, Explain, Elaborate, and Evaluate. Each phase represents a specific stage of the learning process and serves a distinct purpose in helping learners to understand and apply new concepts.

Figure 2.4 illustrates the initial user journey of Beyond using 5E mapping. It contains features that will not be included in the Minimum Viable Product.

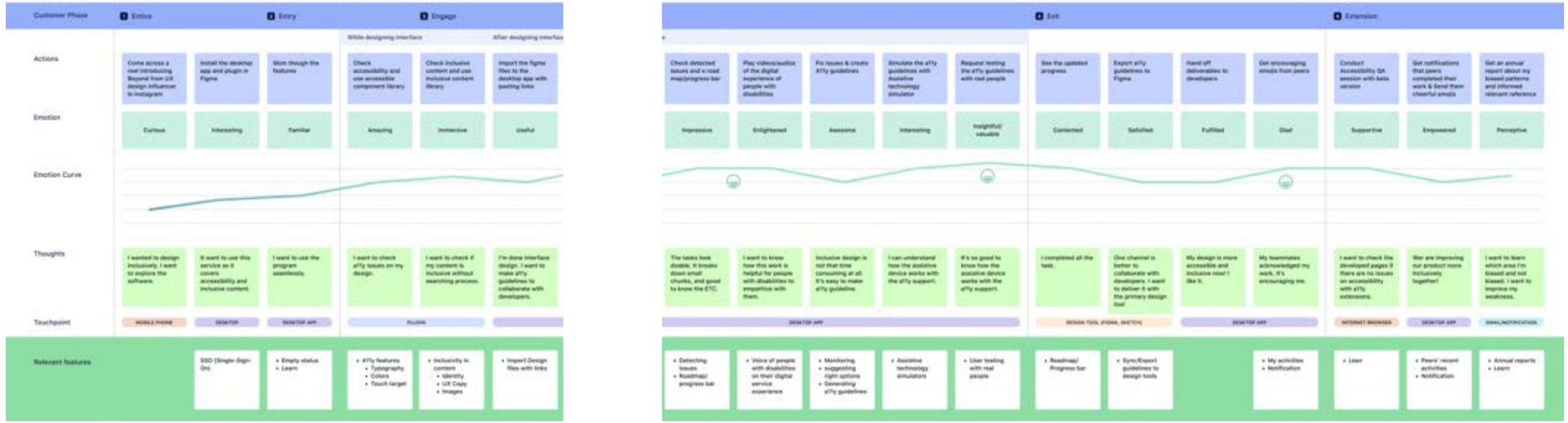


Figure 2.4 User Journey of Beyond

Photo by Young Kim

3.3 Information Architecture

Information architecture refers to the organization and structure of information in a website or other digital product. It involves designing a navigational structure, grouping content into categories and subcategories, and creating a user-friendly interface that allows users to easily find the information they need.

Figure 2.5 illustrates a part of the initial Information Architecure of Beyond. It contains features that will not be included in the Minimum Viable Product.

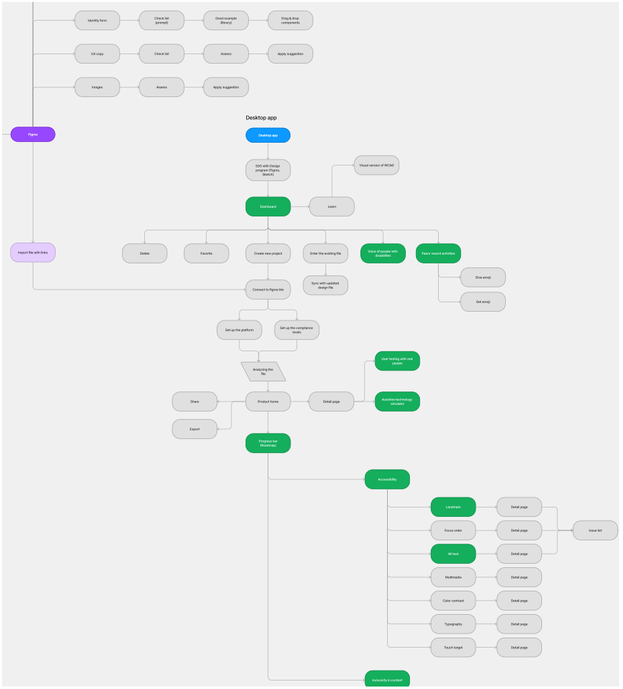
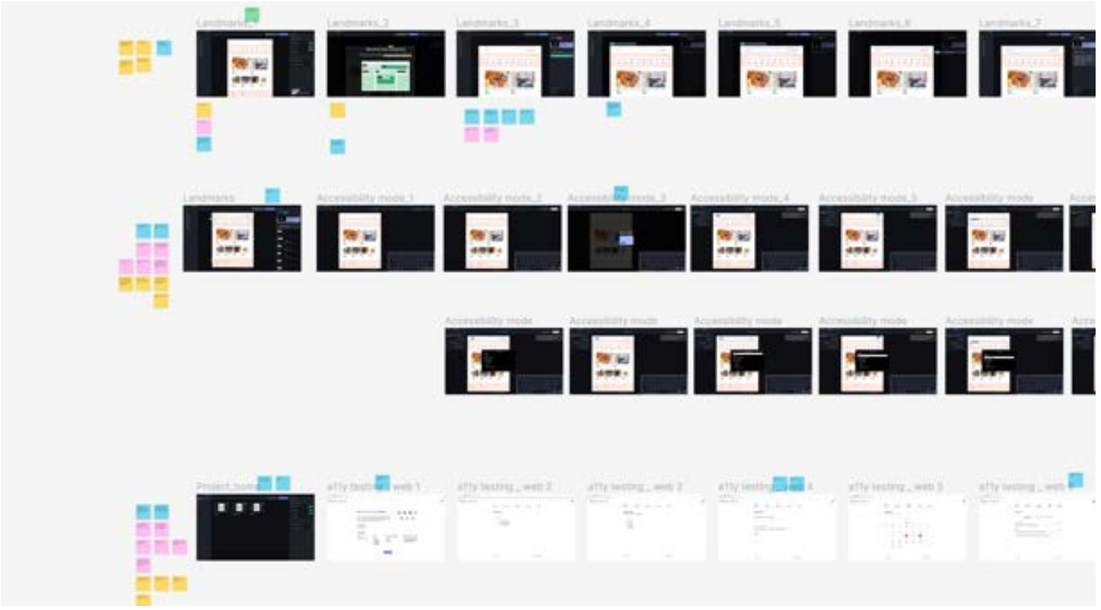


Figure 2.5 A part of Information Architecture of Beyond

3.4 User Testing with Initial Prototypes

User testing with initial prototypes is a critical part of the thesis project process. During this phase, the initial prototypes were tested with four potential users to gather feedback and insights that can be used to improve and refine the design.



Feedback from Testing

"I feel somewhat confident in creating landmarks after using your feature."

"I wasn't sure about the difference between 'complementary', 'aside', and 'region'."

"Being able to interact with prototypes without coding is cool."

"It was straightforward."

"The video would be helpful for learning it. I would like to watch it first."

"Interface design is good"

"It is difficult to distinguish between accessibility mode and accessibility testing. Tooltips would be helpful in differentiating them."

"Accessibility testing is a very unique and valuable process. Rather than relying on accessibility mode, I prefer to test my design with people who have disabilities."

"Accessibility mode looks like the most useful feature. Being able to interact with prototypes without coding is cool. It's also good to be able to check accessibility."

"Creating accessibility annotations was the most useful."

"I expected to see real-time testing. At the very least, I want to see a recording."

"I doubt if the testers are real, and I wonder how the company compensates them."

"I wonder, what if there are many pages to test? It seems that it would not be easy to use this interface for that."

"Good examples are very helpful in helping us understand the usage of 'aria-label'. It makes more sense now."

"I forgot how I did the tutorial. I need more guidance."

"I want to see the action dialog in the selected layer."

"'What kind of role' doesn't sound clear, but I can guess what they mean with the 'primary' and 'secondary' buttons."

3.5 Key Concepts

I. Barrier Awareness

Beyond encourages designers to adopt accessible and inclusive design by actively listening to and learning from the narratives of barriers and challenges faced by marginalized individuals.

[Features]
• Voice of People

II. Annotation Assistant

It is standalone desktop app. It assists designers in adding accurate accessibility annotations to their designs. Annotations help to translate the visual user interface into a linear experience for keyboard and screen reader users. These annotations include focus order, landmarks, and alternative text.

The app also provides the ability to verify their accuracy and suitability within assistive technology environments.

Moreover, the application offers a simple and direct process for unmoderated accessibility testing with individuals who have disabilities. This feature allows designers to receive feedback on the accessibility of their designs from the people who will use them, improving the overall user experience.

[Features]
• Accessibility Annotations
• Accessibility Mode
• Accessibility Testing

III. Interface Design Assistant

This is designed as a plugin for Figma. It includes Automated audits and UI component libraries. Automated audits identity inaccessible visual attributes and exclusive content. The UI component libraries provide the latest and most inclusive forms, allowing designers to use them when they design interface in Figma.

[Features]
• Automated Audits
• UI Component Libraries

Empathize
Learn
Practice

3.6 Key Features and Content

3.6.1 Barrier Awareness

[Voice of People]

Goal

Empathize people with diabilities and marginalized individuals

Painpoints

Not easy to have a chance to listen to barriers and challenges of the assistive technology users and maginalized individuals

Solutions

Provide easy access to narratives of task-related challenges when users enter the task screen

Impact

Motivate designers to practice inclusive and accessible design

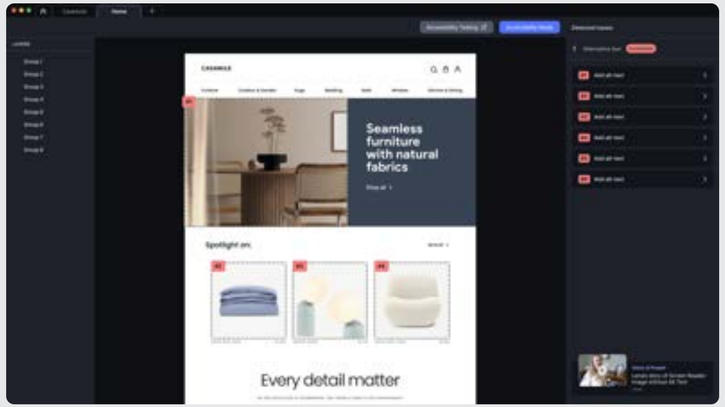


Figure 2.5 illustrates that users can access a video of “Voice of People” when they practice tasks.

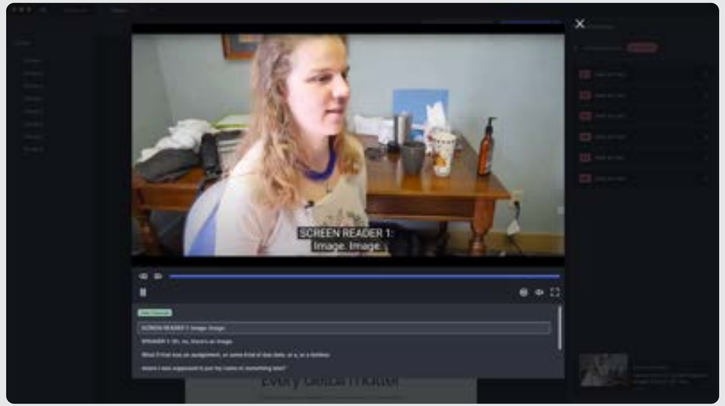


Figure 2.6 shows how visually impaired people struggle with navigating inaccessible websites, presented through narrative play. (Youtube video by ATRC CSU)

3.6.2 Interface Design Assistant

[Accessibility Audit]

Goal

Create accessible visual elements

Painpoints

- Time-consuming to learn the accessibility criterias
- Manual auditing

Solutions

- Detect inaccessible visual attributes automatically
- Suggest accessbile alternatives

Feature list

Color contrast, Typography, Touch target

Impact

Improve accessibility for the low-vision people

[Color Contrast]

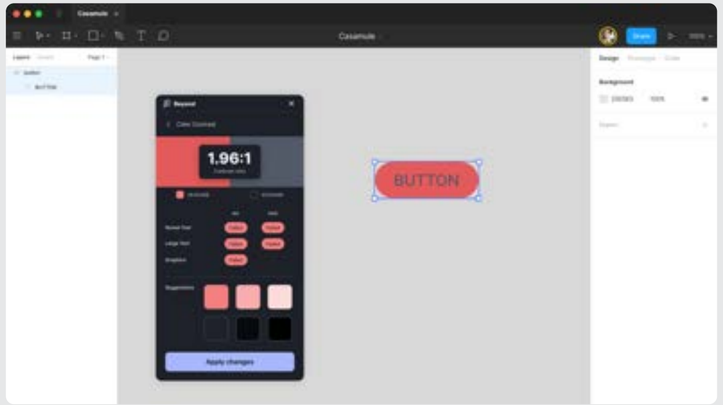


Figure 2.7 shows that Beyond in Figma identifys an accessibility issue with color contrast.

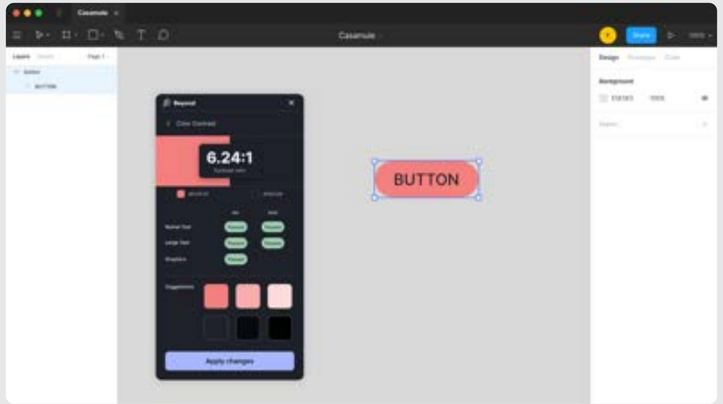


Figure 2.8 shows the accessible alternatives suggested by Beyond.

[Inclusivity Audit]

Goal

Create inclusive content

Painpoints

- Difficult to identify and eliminate exclusive expressions

Solutions

- Identify exclusive content automatically
- Suggest inclusive alternatives

Feature list

UX Copy, Images of People

Impact

Lead to a more positive user experience and a greater sense of belonging for marginalized individuals

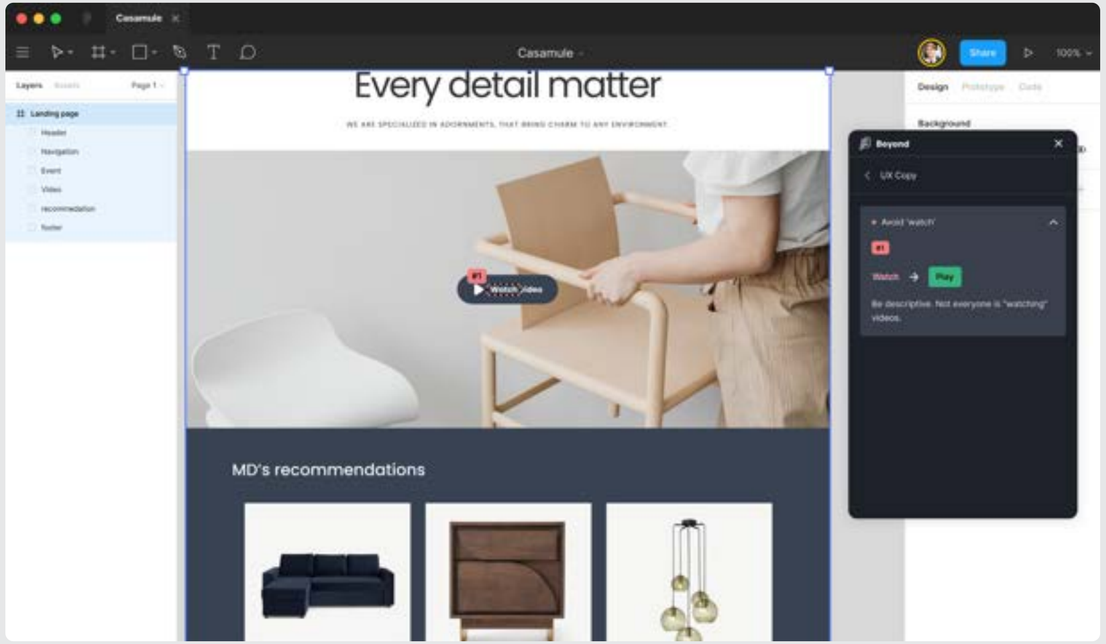


Figure 2.9 illustrates how Beyond identifies exclusive expressions and suggests inclusive alternatives.

[Inclusivity Library]

Goal

Create inclusive identity forms

Painpoints

- Difficult to identify exclusive identity forms
- Burdensome to search for inclusive identity forms

Solutions

Offer a comprehensive library of the latest and most up-to-date inclusive identity forms

Feature List

Pronoun, Gender Identity, Assigned Sex at Birth, Name, Race, Maritial Status

Impact

Lead to a more positive user experience and a greater sense of belonging for marginalized individuals

[Gender Identity Form]

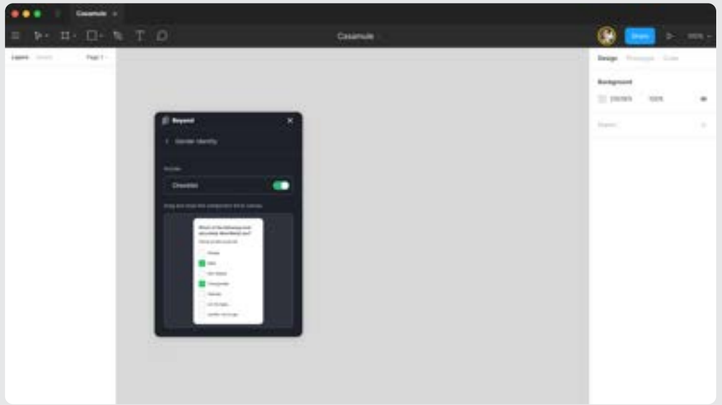


Figure 2.10 illustrates how Beyond in Figma provides an inclusive gender identity form.

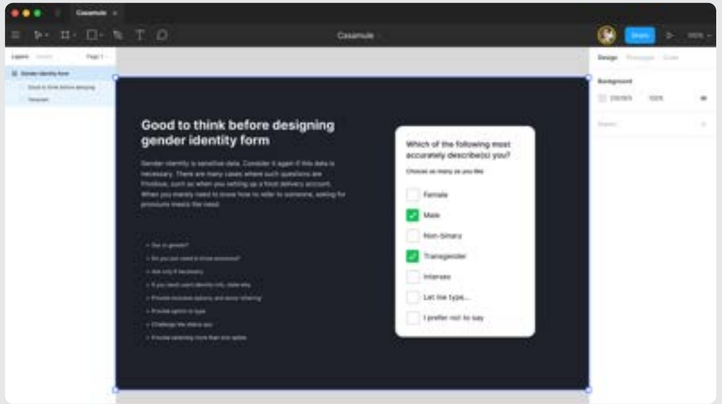


Figure 2.11 displays the checklist that should be used before designing a gender identity form that comes with an inclusive gender identity form.

3.6.3 Annotation Assistant

[Accessibility Annotations]

Goal

Create annotations for accessiblity

Painpoints

The learning curve for WCAG guidelines is steep

Solutions

Integrate annotation creation with education by providing 5-minute learning videos, checklists, good examples, demos, and contextual tutorials

Feature List

Alternative text, focus order, and landmarks

Impact

Improve accessibility for people who rely on assistive technology

[Landmarks]

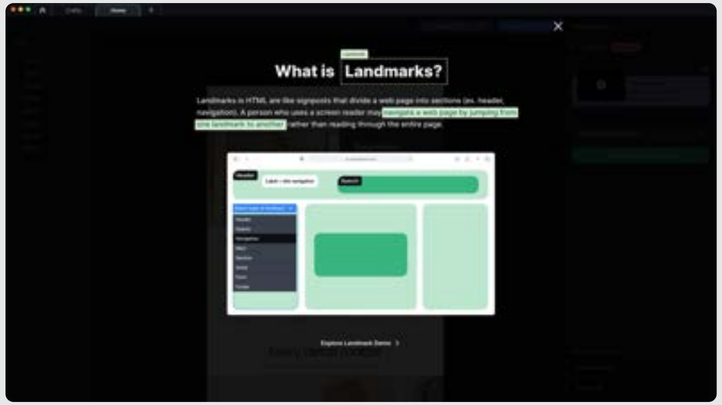


Figure 2.12 illustrates how Beyond educates about what a landmark is with simplified animated guidelines.

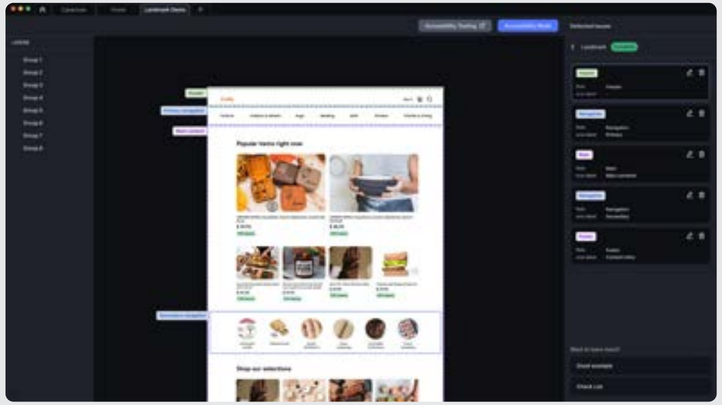


Figure 2.13 illustrates how Beyond educates to create good landmarks with a demo.

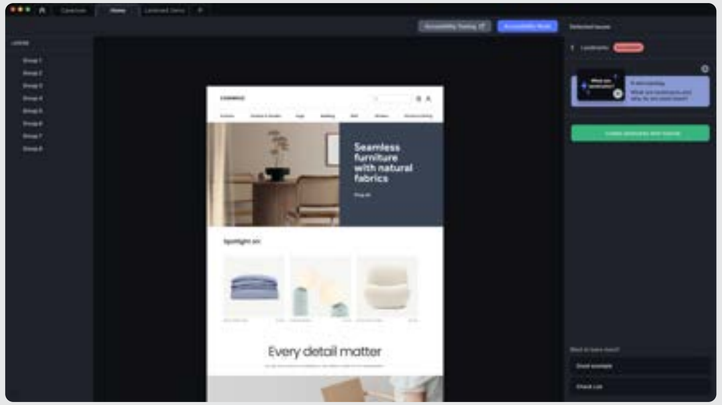


Figure 2.14 illustrates how Beyond supports education with a 5-minute learning video.

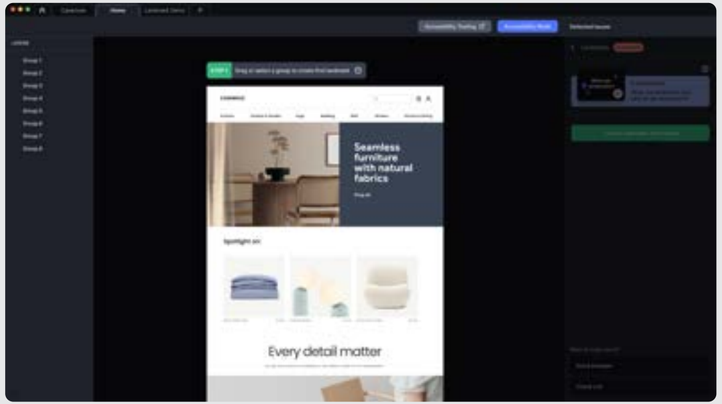


Figure 2.15 illustrates how Beyond supports education with a contextual tutorial (Step 1).

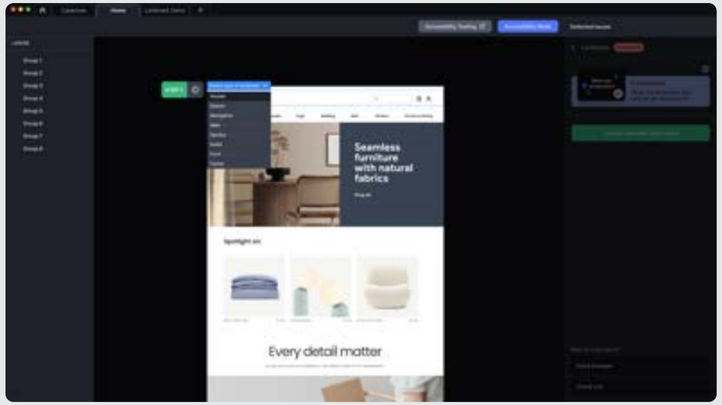


Figure 2.16 illustrates how Beyond supports education with a contextual tutorial (Step 2).

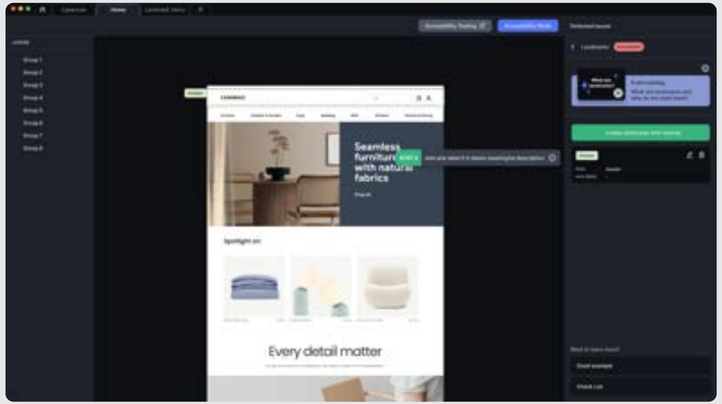


Figure 2.17 illustrates how Beyond supports education with a contextual tutorial (Step 3).

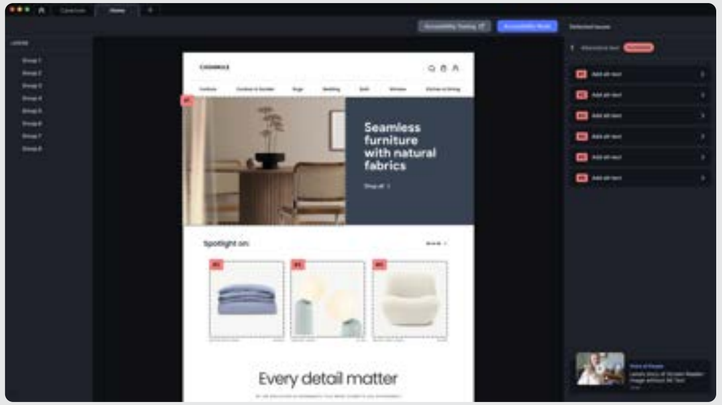


Figure 2.18 illustrates how Beyond detects images that require alternative text.

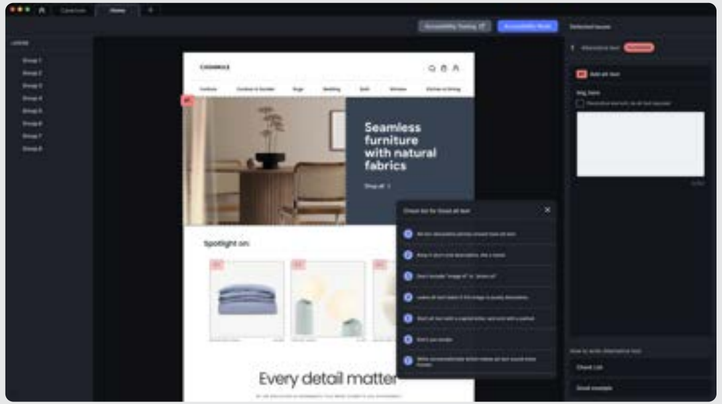


Figure 2.19 shows illustrates how Beyond supports education with a checklist for good alternative text.

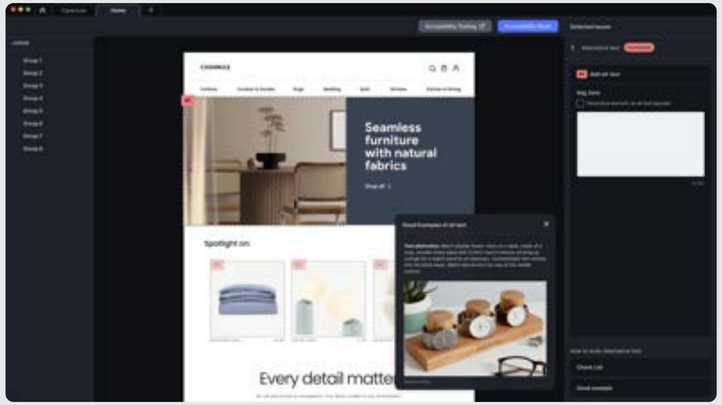


Figure 2.20 illustrates how Beyond supports education with a good examples of alt text. (Source: Etsy)

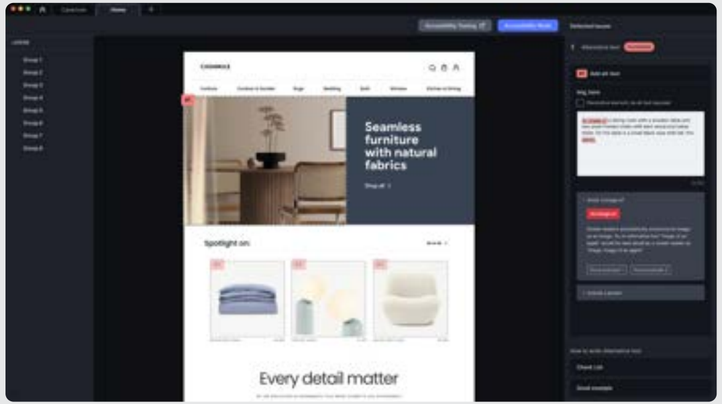


Figure 2.21 illustrates how Beyond help writing alt text in real time by detecting issues and suggesting corrections.

[Accessibility Mode]

Goal

Check annotations for accuracy and suitability

Painpoints

- Lack of an effective way to check the annotations for accessibility
- Lack of familiarity with keyboard navigation

Solutions

- Allow users to simulate annotations within keyboard navigation and screen reader
- Provide a guide to navigate prototypes using a keyboard

Impact

Improve the quality of annotations

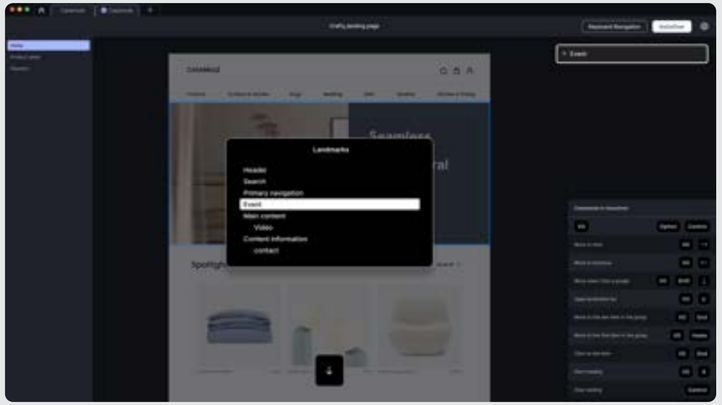


Figure 2.22 shows how Accessibility mode checks annotations for accuracy and suitability using a keyboard and screen reader environment. (Landmarks)

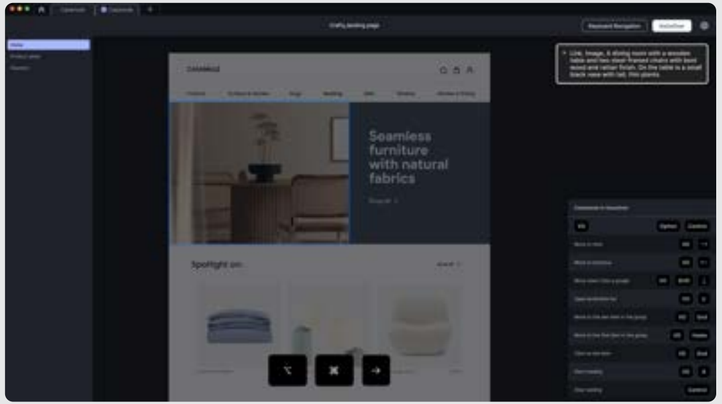


Figure 2.23 shows how Accessibility mode checks annotations for accuracy and suitability using a keyboard and screen reader environment. (Alt text)

[Accessibility Testing]

Goal

Verify accessibility of annotations with people who has disabilities

Painpoints

- Slow recruiting process for people with disabilities as testers
- Train is needed in conducting accessibility testing with people with disabilities

Solutions

- Straightforward process to register unmoderated accessibility testing
- Expedite recruiting process

Impact

Achieve true accessibility

Business Plan

- Involve hiring people with disabilities as testers to expedite the recruiting process and streamline accessibility testing
- Focus on the core industries in the initial stages and plan to expand into other industries
- Provide support to simplify the testing process for individuals with disabilities, ensuring that they are able to participate fully and effectively

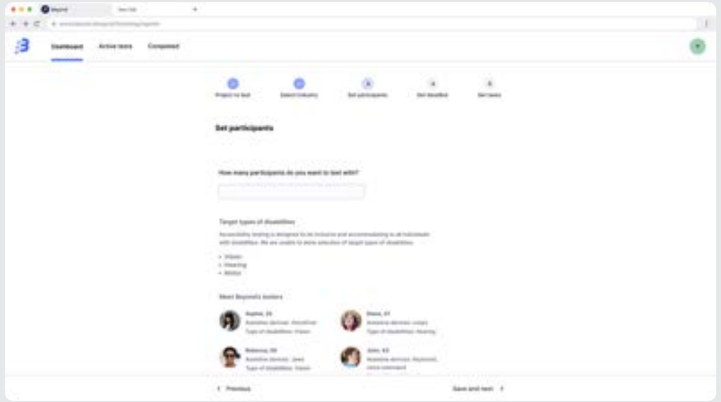


Figure 2.24 illustrates the 5 steps to register for unmoderated accessibility testing with people with disabilities.

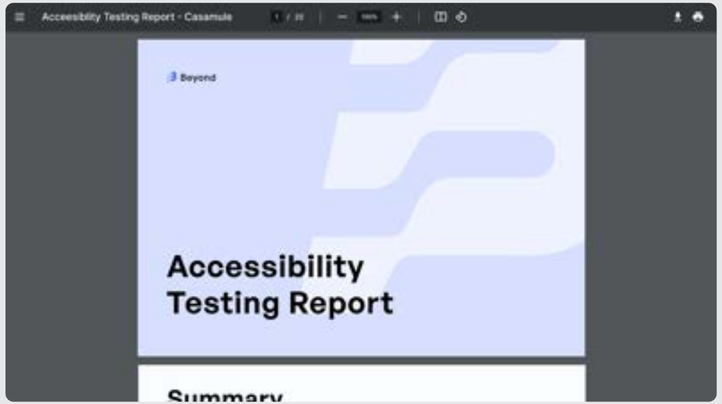


Figure 2.25 illustrates an example of testing result.

CONCLUSION

4.1 Final Thoughts
4.2 Future Work

4 CONCLUSION

4.1 Final Thoughts

Although I have developed a concept tool to streamline the process of inclusive and accessible design, it remains important for designers to dedicate time to learning and practicing these methods. While the tool can certainly assist in the design process, it cannot replace the need for ongoing education and training in inclusive and accessible design principles. Therefore, it is essential for designers to prioritize these skills in order to create truly inclusive and accessible digital experiences.

During the course of my thesis project, the rise of AI technology prompted me to consider incorporating it into my work to automatically generate annotations and alt text for inclusive and accessible design. However, after careful consideration, I decided against utilizing AI for this purpose. While AI may be able to generate alt text and annotations quickly in the near future, the true purpose of my project is to educate designers on inclusive and accessible design principles. Therefore, I chose to focus on education and training for designers, emphasizing the importance of thoughtful and intentional design practices for inclusive and accessible design.

4.2 Future Work

As part of future work, it is important to develop a solid business plan and further refine the features of my thesis project to address the challenges of accessibility testing with people with disabilities. While my thesis project initiated to promote the adoption of inclusive and accessible design principles throughout the traditional design process, from the creation of visual elements to accessibility testing with people with disabilities, there are still limitations to the feasibility of testing with people with disabilities. Through my research, I have identified the challenges that designers face in this area, and there is a need for continued development to make accessibility testing with people with disabilities more feasible.

In order to achieve this, future work may include the refinement of existing tools and resources, as well as a well-defined business plan. By addressing these challenges, we can continue to promote inclusive and accessible design and create more welcoming and accessible digital environments for all users.

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Arusa | Ecommerce UI KIT Template created by Agatha Sakowicz

Product images by Esty

Profile photos by Freepik

I see incredible potential in Young's thesis product, Beyond. Accessibility apps and plugins in the current market have allowed us to understand usability issues through the lens of wider audiences; however, more often than not, they only can provide a limited amount of help around technicality. Beyond goes one step further than that. Its inclusive language feature can help designers to contextualize the importance of seeing ALL users in broad spectrums. It ultimately allows us to create products that invite users and enables us to avoid languages that could cause emotional barriers for the users. With Beyond's step-by-step review feature, I'm sure the designers will understand the logic behind each suggestion, educating themselves along the way.

I am genuinely impressed with Young's imagination and intention behind the product. Although Beyond is a thesis idea, I believe it would do well in the competitive market as a fully actualized product with the right business plan.

Sara Bang, Senior UX Designer

TBD

Elyse Voegeli, Thesis advisor/ UX Designer + Researcher

Beyond

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